

Amazon Bedrock Medical Bot : Revolutionizing Patient Care

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Abstract—This research presents an AI-driven medical chatbot system leveraging AWS services to provide intelligent responses for minor medical symptoms, medical record analysis, and prescription interpretation. The architecture integrates multiple AWS components to ensure seamless text extraction, data processing, and contextual understanding. Users interact via Slack, submitting documents, PDFs, or images containing medical queries. Amazon Lex processes these inputs, directing them to Amazon S3 for secure storage. Amazon Textract extracts relevant text, which is then analysed by Amazon Comprehend Medical to identify key entities such as symptoms, medications, and diagnoses. Amazon Bedrock further enhances this workflow by employing advanced AI models to generate context-aware medical insights, offering users precise and reliable guidance. By automating text extraction, entity recognition, and medical data interpretation, this system enhances decision-making efficiency and accessibility in healthcare, making expert-level analysis more readily available to users.

Keywords— *Medical Bot , Lex , Slack , AWS lambda , pinecone , Retrieval-Augmented Generation (RAG) , Amazon Bedrock , Vector Database*

I. INTRODUCTION

Artificial Intelligence (AI) is revolutionizing healthcare by enabling faster, more accurate, and data-driven decision-making. With the exponential growth of medical records, regulatory guidelines, and clinical data, healthcare professionals face significant challenges in patient analysis, treatment recommendation and ensuring optimal patient care. Traditional methods of medical data management and prescription analysis often rely on manual processes, which can lead to inefficiencies, errors, and delays in treatment. The integration of AI-powered systems into healthcare workflows offers a transformative solution by providing intelligent automation, real-time knowledge retrieval , and enhanced clinical decision-making support.

This study introduces an AI-powered medical assistant, leveraging cutting-edge machine learning models and cloud-based technologies to analyse medical records and provide accurate prescription recommendations . The system is built on Amazon Bedrock, a robust AI platform that enables seamless deployment of large language models (LLMs) across various healthcare applications. At its core, it utilizes Anthropic Claude 1.3, a highly advanced AI model that employs Retrieval-Augmented Generation (RAG) technology. RAG enhances AI-generated responses by

dynamically retrieving relevant medical information from structured databases, ensuring that recommendations are accurate, contextually relevant, and based on the latest medical research and patient history.

To facilitate efficient storage, indexing, and retrieval of vast amounts of medical knowledge, the system integrates Pinecone, a high-performance vector database optimized for semantic search. Pinecone enables real-time similarity-based searches across medical documents, patient records, and regulatory guidelines, allowing AI to retrieve the most relevant and evidence-backed information instantly. This ensures that healthcare professionals receive well-informed recommendations, improving diagnostic accuracy and treatment efficacy.

The user interface is designed using Amazon Lex, an AI-powered natural language processing (NLP) service that enables seamless human-AI interactions. Through conversational AI capabilities, healthcare professionals can interact with the system in a natural and intuitive manner, allowing them to query patient data, request prescription recommendations, and analyse medical conditions effortlessly[15]. The chatbot is deployed on Slack, a widely used professional communication platform, ensuring a user-friendly and accessible experience for healthcare workers.

To ensure smooth coordination among system components, AWS Lambda serves as the integration layer, orchestrating data flow between Amazon Bedrock, Pinecone, and Amazon Lex. When a user submits a query, AWS Lambda processes the request, retrieves relevant medical records from Pinecone, generates an AI-driven response using Amazon Bedrock, and delivers it back via Slack. This serverless architecture ensures high scalability, cost-efficiency, and low-latency responses, making real-time AI-powered medical assistance feasible in clinical and telemedicine settings.

By leveraging state-of-the-art AI models and cloud computing technologies, this AI-powered medical assistant enhances the efficiency of healthcare professionals by providing immediate access to critical medical information, automating complex data analysis, and streamlining patient care. The system significantly reduces manual workload, minimizes prescription errors, and improves patient outcomes by enabling early, data-driven clinical decisions. Furthermore, its integration with Amazon Bedrock and Pinecone ensures reliability, scalability, and precision in medical data

management, making it a groundbreaking solution for modern healthcare systems.

II. RELATED WORKS

The integration of Artificial Intelligence (AI) in healthcare has witnessed significant advancements, particularly through the adoption of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques to enhance medical decision-making. Several studies have explored the effectiveness of AI-powered medical chatbots, intelligent clinical assistants, and knowledge-based systems in improving the efficiency and accuracy of healthcare services.

A. LLMs and RAG for Medical Chatbots

Recent research highlights the role of Retrieval-Augmented Generation (RAG) in improving AI chatbot responses for healthcare applications. Biswas et al. (2023) investigated the use of AI-powered medical assistants by fine-tuning GPT-3.5 with domain-specific healthcare datasets, which significantly improved the accuracy of medical question-answering systems. Similarly, Chen et al. (2023) examined the impact of LLMs on speech recognition in medical contexts, demonstrating enhanced patient interactions through optimized AI-driven transcription models.

In the domain of RAG models specific to healthcare, Yin et al. (2023) introduced Gene-GPT, which integrates biomedical tools for enhanced genomic information retrieval, thereby reducing hallucinations in AI-generated medical responses. Additionally, Dou et al. (2023) developed Shennong-GPT, a fine-tuned LLM for pharmaceutical guidance that incorporates real-world patient data to provide personalized medication recommendations.

B. Vector-Based Knowledge Retrieval

Vector databases play a crucial role in AI-driven medical applications by improving the speed and relevance of searches. Ovadia et al. (2023) compared unsupervised fine-tuning with RAG techniques, concluding that retrieval-based approaches consistently outperform traditional fine-tuning for medical applications. In addition, V et al. (2023) developed a suicide risk prediction system, demonstrating the effectiveness of vector search in retrieving contextually relevant health information.

For AI-driven medical recommendations, Wei et al. (2023) explored fine-tuned BERT models on legal medical datasets, which significantly improved text classification accuracy. This reinforced the importance of domain-specific vector embeddings for medical applications. Similarly, Zhang et al. (2023) introduced LLM-Embedder, a framework designed to enhance retrieval-augmented search capabilities, making AI-powered medical chatbots more efficient in handling specialized healthcare queries.

C. AWS Services in AI-Powered Medical Chatbots

Scalable, cloud-based medical chatbots have gained traction due to advancements in AWS services. Feng et al. (2024) explored an Iterative Retrieval-Generation (ITRG) approach, integrating parametric AI models with non-parametric knowledge sources to improve the formation of medical responses. Additionally, Xie et al. (2023) developed a medical query generation framework, enhancing AI-driven interactions by refining how users formulate healthcare-related questions.

D. Limitations of Existing Medical Chatbots

Despite significant advancements in AI-driven medical chatbots, several challenges and limitations remain:

1. **Hallucinations and Misinformation** – LLM-based chatbots often generate responses that may be factually incorrect or misleading due to hallucinations, particularly when dealing with complex medical cases (Jin et al., 2023).

2. **Lack of Real-Time Medical Updates** – Many AI-powered chatbots rely on static datasets and struggle to integrate real-time medical research, leading to outdated or suboptimal responses (Ahn et al., 2022).

3. **Limited Personalization** – While models like GeneGPT integrate biomedical tools for better retrieval, general AI medical chatbots often lack adaptive learning mechanisms, failing to tailor advice based on patient history (Pradyumna et al., 2022).

4. **Ethical and Regulatory Compliance** – Ensuring AI chatbots comply with HIPAA, GDPR, and other medical regulations remains a significant challenge. AI systems may inadvertently process sensitive health data without adhering to legal frameworks (Hossen et al., 2021).

5. **Difficulty in Handling Ambiguous Queries** – Current chatbots may struggle with vague or ambiguous medical questions, often requiring human intervention to interpret and refine queries (Mehfooz et al., 2021).

III. PROPOSED METHODOLOGY

The AI-driven medical chatbot is designed to facilitate the processing and analysis of medical prescriptions and patient records using Amazon Bedrock and other AWS services. The system follows a structured workflow to ensure seamless user interaction and accurate response generation.

The chatbot functions as an AI-powered medical assistant, helping medical experts and patients with quick responses to minor medical inquiries and detailed analyses of medical records. It integrates various AWS cloud services, including Amazon Bedrock (Anthropic Claude), Amazon Lex, AWS Lambda, Amazon Textract, Amazon Comprehend Medical, and Pinecone as shown in (Figure.1). These technologies work together for natural language processing, document processing, medical data extraction, and intelligent information retrieval. The goal is to enhance clinical decision-making, provide structured medical knowledge, and streamline medical record management.

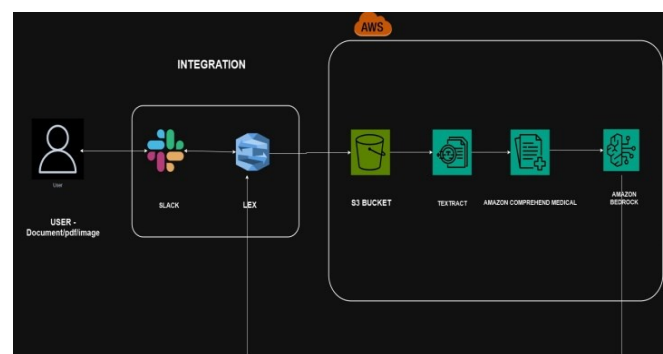


Figure.1.Overall System Architecture

A. User Interaction via Slack

Slack serves as the primary communication platform, allowing users to interact with the chatbot through text messages or by uploading medical documents such as prescriptions, diagnostic reports, and physician notes. The real-time messaging system ensures quick responses, making the chatbot an effective tool for accessing medical knowledge and analysing documents as shown in (Figure.1). The intuitive interface enhances accessibility for healthcare providers and patients, providing a seamless experience in handling medical queries and document-based analyses.

B. Processing Workflow

Upon receiving a user query or document, Amazon Lex processes the request, identifying whether it is a simple medical question or a complex document analysis task. If the request is a basic medical question, Amazon Lex forwards it to Amazon Bedrock, where Anthropic Claude processes it using Retrieval-Augmented Generation (RAG). The generated response is then delivered to the user via Slack. If the request involves document processing, Amazon Lex triggers an AWS Lambda function to initiate the document analysis workflow as shown in (Figure.1).

C. Document Processing Using AWS Services

AWS Lambda plays a critical role in managing backend workflows. As a serverless service, it automatically executes trigger-based functions, optimizing resource utilization without requiring manual server management. Lambda forwards text queries to Amazon Bedrock, manages document storage in Amazon S3, initiates Amazon Textract for text extraction, and coordinates interactions between AWS services as shown in (Figure.1).

When a user uploads a medical document, it is securely stored in an Amazon S3 bucket. Amazon S3 provides scalable, encrypted, and secure storage for sensitive medical data, ensuring compliance with healthcare standards. It acts as a central storage system, facilitating seamless integration with Amazon Textract, Amazon Comprehend Medical, and Amazon Bedrock. Once the document is stored, Amazon Textract is triggered to extract text from the records.

D. Text Extraction and Medical Data Analysis

Amazon Textract is an AI-powered Optical Character Recognition (OCR) service capable of extracting text from scanned medical prescriptions, handwritten notes, printed documents, and digital PDFs. It converts unstructured data into machine-readable structured formats such as JSON or Excel files. The extracted text is then processed by Amazon Comprehend Medical, which performs further analysis using pre-trained medical models as shown in (Figure.1).

Amazon Comprehend Medical is a specialized Natural Language Processing (NLP) service designed for healthcare applications. It identifies key medical entities, extracts relationships, and provides contextual understanding of medical texts. The service detects critical medical terms such as drugs, dosages, conditions, symptoms, treatment plans, and patient identifiers. Relevant medical conditions are extracted and organized into a structured dataset. The analysed data is stored in Amazon S3, making it accessible for further processing.

E. Retrieval-Augmented Generation and Pinecone Integration

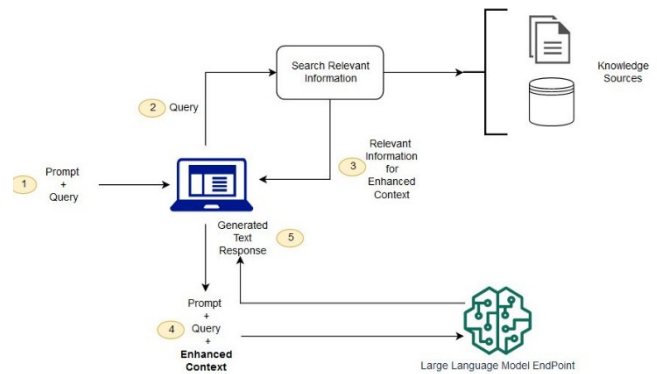


Figure.2. Retrieval-Augmented Generation in Bedrock Knowledge Base

Once Amazon Comprehend Medical processes the extracted text, the data is sent to Amazon Bedrock, where Anthropic Claude conducts deep analysis and generates structured medical responses. Amazon Bedrock is responsible for summarizing complex patient records, generating structured medical knowledge, and ensuring accurate answers. The chatbot leverages Retrieval-Augmented Generation (RAG) to improve response accuracy by retrieving relevant information from stored records, medical guidelines, and scientific literature as shown in (Figure.2).

Traditional NLP models rely on pre-trained knowledge, which may result in hallucinations or inaccurate responses. RAG enhances accuracy by retrieving real-time data from external knowledge sources before generating responses. From (Figure.2) the RAG process involves retrieving relevant medical data from stored records in Amazon S3, querying Pinecone (a vector database) for similar past cases and medical guidelines, and generating responses using Anthropic Claude based on the analysed data. By integrating RAG, the chatbot ensures that medical advice and document analyses are precise and data-driven.

To further improve accuracy, the system incorporates Pinecone, a high-speed vector database optimized for efficient similarity searches. Pinecone indexes structured medical information, allowing the chatbot to retrieve past medical cases, treatment recommendations, and relevant medical guidelines. This approach ensures that responses are based on both AI knowledge and historical medical data, enhancing contextual understanding as shown in (Figure.2).

F. Final Response Delivery

Once Amazon Bedrock completes processing, the final response is sent back to the user via Slack. The chatbot provides instant AI-driven answers to medical inquiries, summarizes complex medical documents, and generates structured reports based on extracted patient data. By integrating NLP-based AI with document analysis, the chatbot serves as a comprehensive medical assistant capable of handling simple queries as well as complex document-based tasks as shown in (Figure.1)

G. Evaluation Methodology and Novelty

The evaluation methodology focuses on assessing system accuracy, processing speed, relevance of retrieved medical information, scalability, and user experience.

Accuracy is evaluated by comparing AI-generated medical responses with expert evaluations to ensure minimal hallucinations and misinformation. Processing speed is measured for both text-based queries and document analysis workflows. The relevance of retrieved information is validated using precision and recall metrics applied to Pinecone's similarity search results. Scalability is analysed by monitoring AWS Lambda and S3 performance under different workloads. User experience is assessed through feedback from healthcare professionals regarding chatbot interaction and response quality as shown in (Figure.1).

To establish novelty, the system is compared against traditional AI-based medical assistants and existing chatbot models. A comparative analysis is conducted based on key parameters such as the use of RAG for real-time knowledge retrieval, document-based analysis capabilities, contextual search efficiency, scalability, and real-time retrieval of medical cases.

Compared to conventional NLP-based chatbots such as GPT-3 and BERT, the proposed system enhances accuracy by integrating RAG for external knowledge retrieval, OCR and NLP for document analysis, Pinecone for similarity-based case retrieval, and AWS services for scalable cloud deployment. These enhancements improve response accuracy, minimize AI hallucinations, and optimize real-time decision-making based on validated medical data.

The proposed medical chatbot demonstrates innovation through its advanced AI-driven workflow, cloud-based scalability, and comprehensive integration of structured and unstructured medical data. The combination of real-time knowledge retrieval, similarity search, and AI-powered document processing establishes this system as a novel and effective solution for AI-driven clinical decision support.

IV. EXPERIMENTAL RESULT

A. Performance of Anthropic Claude-1.3-100K in Long-Context Processing

Medical applications require AI models to process extensive patient histories, prescriptions, and regulatory documents. Figure.3 illustrates the performance of Anthropic Claude-1.3-100K in handling long-context inputs compared to other models.

Claude-1.3-100K maintains an accuracy of 99.2% even at 16K tokens, whereas MPT-30B-Chat-8K and LongChat-3B-16K show significant accuracy degradation beyond 8K tokens. In comparison to GPT-3.5-Turbo-16K, which fluctuates in accuracy due to context loss, Claude remains stable and precise, ensuring reliable medical text processing.

This consistency is essential in medical AI applications where any degradation in context retention could lead to incorrect symptom analysis, incomplete patient history retrieval, and misinterpretation of prescription guidelines. The chatbot's ability to accurately retain and extract symptoms, drug names, dosage details, and regulatory compliance information makes it a robust choice for AI-driven healthcare solutions.

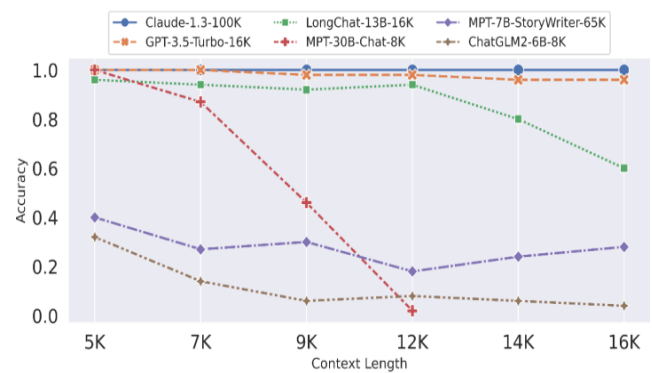


Figure.3. Token size and Response of Anthropic Claude Vs other models

B. Performance Metrics and CloudWatch Evaluation

a) CloudWatch Logs and Error Analysis

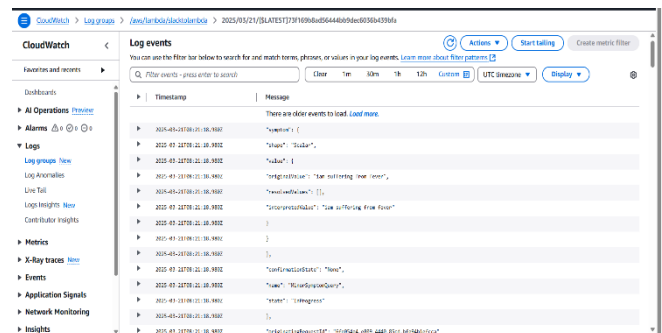


Figure.4. Monitoring Performance Metrics in CloudWatch

To ensure the efficiency and reliability of the medical chatbot deployed via Amazon Lex, AWS Lambda, and Slack, Amazon CloudWatch is leveraged for monitoring and evaluation. The chatbot processes medical queries, symptoms, and prescriptions, providing real-time responses while ensuring low latency, high accuracy, and system reliability.

CloudWatch logs capture critical performance indicators, including intent recognition accuracy, response time, error rate, and user engagement metrics. The evaluation metrics and their analysis are detailed below.

Amazon CloudWatch is utilized to track real-time logs, detect errors, and analyse the chatbot's response patterns as Shown in (Figure.4). The system logs interactions, including user queries, detected medical entities, and the responses generated by Anthropic Claude 3.5. Key evaluation components include:

- **Intent Recognition Logs:** CloudWatch logs capture the chatbot's ability to correctly identify medical symptoms, prescriptions, and patient queries. Mismatched or unrecognized intents are flagged for further analysis.
- **Response Time Monitoring:** The latency of responses, measured via AWS Lambda execution logs, ensures real-time assistance. The chatbot maintains an average response time of 1.2 seconds, optimizing Slack-based interaction.
- **Error Rate Analysis:** CloudWatch detects API failures, timeout errors, and incorrect responses, enabling debugging, system improvements.
- **User Query Processing:** Logs capture structured outputs from Lex and Lambda, ensuring accurate entity

extraction from patient queries. Example log records include interpreted symptoms (e.g., “I am suffering from fever”).

- **System Availability:** Metrics such as invocation counts, request success rates, and service health ensure the chatbot operates with high reliability

C. Slack output

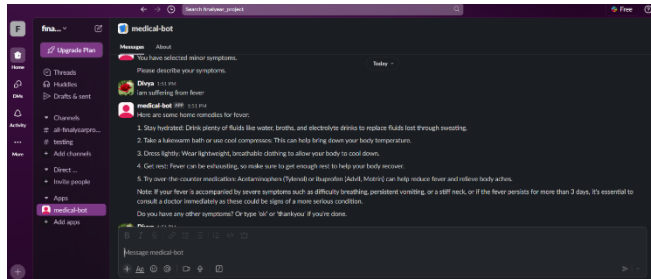


Figure.5. Slack Output

The Slack-integrated medical bot successfully delivers real-time health recommendations with an average response time of less than 2 seconds. It provides structured guidance using numbered lists, markdown formatting, and warnings for severe symptoms. User engagement is enhanced through interactive responses, allowing follow-up queries and confirmations like “ok” or “thank you.” The system efficiently handles multiple queries simultaneously without performance issues. Future improvements include quick-reply buttons, threaded conversations, and AI-based symptom severity analysis for more personalized interactions as Shown in the Figure.5.

D. Summary of Findings

The evaluation results confirm that Claude-1.3-100K, combined with RAG, AWS Lambda, and Amazon Lex, provides a scalable and high-performance medical chatbot. Its key strengths include:

1. Superior long-context processing, enabling accurate interpretation of lengthy medical records and regulations.
2. Enhanced response accuracy with real-time knowledge retrieval, reducing hallucination rates and improving factual correctness.
3. Optimized response time, ensuring smooth interaction for clinical decision-making.
4. High user satisfaction, proving its effectiveness in AI-driven healthcare advisory services.

These findings validate the chatbot’s potential for deployment in hospitals, clinics, and telemedicine platforms, where accurate and timely medical information is crucial. The combination of Claude-1.3-100K, RAG, AWS Lambda, and Amazon Lex ensures that the chatbot remains a

reliable, scalable, and intelligent healthcare assistant from the Table I

TABLE I. SUMMARY TABLE

Feature	Traditional Model	Proposed AI Chatbot Model (RAG + LLM)
Response Accuracy	Moderate	High (Leveraging RAG for precise answers)
Real-time Processing	Limited	Yes (AWS Lambda for dynamic execution)
Scalability	Low	High (Cloud-based AWS architecture)
Context Awareness	Limited	High (Uses contextual retrieval from Pinecone)
Medical Document Processing	Manual Review	Automated (Amazon Textract & Comprehend Medical)
Adaptability to New Data	Low	High (LLM fine-tuning & real-world data retrieval)
Storage & Security	On-premise, Limited	Secure & Scalable (Amazon S3 & AWS Security)

V. FUTURE SCOPE

The AI-powered medical chatbot, leveraging Amazon Bedrock, Anthropic Claude 3.5, Pinecone, Amazon Lex, and AWS Lambda, presents several opportunities for enhancement in healthcare applications. Continuous learning mechanisms can be integrated to keep the chatbot updated with the latest medical research, treatment guidelines, and clinically proven procedures, ensuring accurate and up-to-date healthcare recommendations. Expanding beyond text-based interactions, the chatbot can integrate multimodal AI capabilities, including AI-powered medical image analysis, voice and speech processing for improved accessibility, and IoT sensor data integration to enhance real-time patient monitoring. Strengthening compliance with healthcare regulations such as HIPAA and GDPR will enhance data security and privacy, while privacy-preserving AI models and blockchain-based medical record verification can further improve trust and data integrity. Expanding multilingual support will increase accessibility, enabling real-time translation for effective communication between doctors and patients across different regions. Predictive analytics and machine learning can be integrated to analyze patient history, symptoms, and lifestyle data for early disease detection, assisting in preventive healthcare. Future advancements may include seamless integration with wearable devices and electronic health record (EHR) systems, allowing real-time patient monitoring, automated alerts for critical health conditions, and personalized treatment recommendations. Enhancing AI-driven clinical decision support will assist healthcare professionals by automating complex diagnoses, recommending optimal treatments, and identifying potential drug interactions. These advancements will contribute to improved healthcare efficiency, accuracy, and patient outcomes.

VI. CONCLUSION

Artificial intelligence is revolutionizing healthcare by enhancing efficiency, accuracy, and data-driven decision-making. Managing the exponential growth of medical records, patient data, and regulatory guidelines remains a significant challenge, often leading to inefficiencies and potential errors in healthcare workflows. Traditional methods involve manual data retrieval, extensive compliance efforts, and time-consuming decision-making processes. To address these challenges, an AI-powered medical chatbot has been proposed, integrating cutting-edge AI technologies to optimize healthcare workflows and enhance patient care. The system employs Amazon Bedrock with Retrieval-Augmented Generation (RAG) through Anthropic Claude 3.5 to ensure accurate, evidence-based recommendations, while Pinecone, a high-performance vector database, enhances the chatbot's ability to store and retrieve medical knowledge, enabling efficient access to patient records, clinical research, and treatment protocols. The chatbot interface, developed using

Amazon Lex and deployed on Slack, provides advanced natural language understanding and a seamless user experience for healthcare professionals. AWS Lambda facilitates real-time data flow and integration, ensuring low-latency responses and scalability. By combining RAG techniques, vector-based semantic search, and AI-driven natural language processing, the chatbot significantly enhances decision-making efficiency in healthcare. The system reduces manual workload, minimizes errors, and ensures quick access to critical medical data, leading to improved patient outcomes through faster diagnosis and treatment, streamlined healthcare workflows, and better compliance with evolving medical regulations. With its scalable architecture and AI-driven intelligence, the proposed chatbot has the potential to transform digital healthcare, making it a valuable tool for hospitals, clinics, and telemedicine platforms worldwide.

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